

Andrew Peene

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[Portfolio](#) | Ithaca, NY

EDUCATION

Cornell University, College of Engineering

Bachelor of Science, Mechanical Engineering

GPA: 3.5 | Minor: Aerospace Engineering

[William A. Dart Memorial University Scholarship](#)

Selected as the sole 2024 STEM scholarship recipient from 57 applicants

Expected May 2028

Ithaca, NY

Cayman Prep and High School

A-levels: A* - Math, Computer Science, Business | A – Physics

IGCSE: Honors | 5 A*'s, 5 A's, and a distinction in Additional Math

June 2024

Grand Cayman

Diploma Received June 2022

WORK EXPERIENCE

Crossover Solutions

May 2026 – July 2026

Quality and Process Engineering Intern

Detroit, MI

- Evaluated 12 process failure modes using PFMEA and 8D root-cause analysis, collecting and analyzing production data to identify root causes and develop corrective actions for an EV battery production line.
- Conducted 30+ time and cycle studies across 3 assembly stations, analyzing process data to identify bottlenecks, inefficiencies, and opportunities to improve production throughput.
- Collaborated with PLC, process, and quality engineers to implement 7 corrective actions across 3 manufacturing cells, including Manufacturing Execution System (MES) maintenance alerts for adhesive replacement and process changes to reduce cycle time and production downtime.

Dart Enterprises

June 2025 – August 2025

Design Engineering Intern

Grand Cayman

- Utilized AutoCAD and Revit to support 3D model development and coordination for commercial construction projects with a combined value exceeding \$100 million, assisting with design visualization and constructability review.
- Collaborated with contractors, consultants, and project stakeholders to identify and resolve design discrepancies, helping ensure architectural, structural, and MEP drawings remained coordinated with project requirements.
- Researched and evaluated international architectural firms for future development opportunities, comparing design capabilities, project experience, and suitability for upcoming large-scale projects.

PROJECTS

Suction Device for Removal of Spotted Lanternflies

January 2026 – May 2026

- Evaluated 2 mechanical removal concepts and compared axial and centrifugal airflow configurations, selecting a centrifugal design to direct captured spotted lanternflies into a collection chamber.
- Designed and fabricated 3 prototype iterations using Fusion 360, 3D-printed PLA components, and soldered electronics, delivering a functional prototype for approximately \$100 against a \$300 budget.
- Conducted 3 prototype testing trials using paper insect simulants and cardboard grape clusters, successfully removing simulated spotted lanternflies from vines without disturbing the simulated fruit.

2D Computational Wind Tunnel

August 2026

- Developed an interactive 2D fluid-flow simulation in Python using the D2Q9 Lattice Boltzmann Method to reinforce introductory fluid mechanics concepts while gaining hands-on experience with numerical simulation.
- Implemented multiple geometries, adjustable angle of attack, real-time velocity-field visualization, and qualitative lift/drag indicators to compare how object geometry and orientation influence surrounding flow behavior.
- Improved simulation performance and numerical stability through iterative debugging, code optimization, and validation of expected flow behavior, refining both the numerical implementation and responsiveness of the application.

ADDITIONAL

Technical: Fusion 360 | MATLAB | AutoCAD | Python | Revit | Simulink | PFMEA | Prototyping | Power BI

Certifications: FAA Private Pilot's License | PADI Divemaster

Interests: Cornell Club Hockey D2 | Cornell Poker Club | Spearfishing | Surfing | Skiing